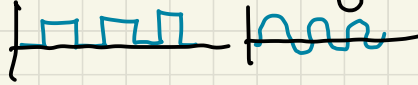


2059K

Devreler

Google Classroom = bljvcr

İşaretler → Dijital ve Analog



4 bit = nibble

2 byte = word

$$a = \frac{V_{\max}}{2^n - 1}$$

↓
çözünürlük

↓
bit adedi

~~ÖR~~ 4 bitlik bir DAC devresinin çıkış gerilim aralığı 0V ile 6V arasında. 0000 sayısal girişi 0V'a, 1111 6V'a karşılık geliyor

$$\begin{array}{c|c} 0000 & 0V \\ 0001 & ? \\ 0010 & 0.8 \end{array} \rightarrow a = \frac{6V}{2^4 - 1} = \frac{6}{15} = 0.4$$

0.8 → 0.4 a = 0.4 → çözünürlük

Deramında binary to decimal, hexadecimal to decimal örnekleri çözdük

~~ÖR~~ $(33)_{10} = (?)_2$

$$\begin{array}{r} 33 \div 2 = 16 \text{ R } 1 \\ 16 \div 2 = 8 \text{ R } 0 \\ 8 \div 2 = 4 \text{ R } 0 \\ 4 \div 2 = 2 \text{ R } 0 \\ 2 \div 2 = 1 \text{ R } 0 \\ 1 \div 2 = 0 \text{ R } 1 \end{array}$$

100001

LSB

MSB

~~ÖR~~ $(153.51)_8 = (?)_{10}$

$$\begin{array}{c} 153.51_8 \\ \downarrow \downarrow \downarrow \downarrow \downarrow \\ 8^2 \quad 8^1 \quad 8^0 \quad 8^{-1} \quad 8^{-2} \\ \downarrow \downarrow \downarrow \downarrow \downarrow \\ 64 \times 1 + 8 \times 5 + 1 \times 3 = 107. \quad \frac{51}{64} \end{array}$$

~~ÖR~~ $(100111000011)_2 = (?)_{16}$

$$\begin{array}{c} 1001 \quad 1100 \quad 0011 \\ \downarrow \quad \downarrow \quad \downarrow \\ 9 \quad 12 \quad 3 \\ \downarrow \quad \downarrow \\ 12 = C \end{array}$$

9C31₁₆

Çıkarma işleminde

2's complement kullanılır

ters çevir +1 ekle

2's complement ÖR

$$\begin{array}{r} 10011 \\ - 10100 \\ \hline 00001 \end{array}$$



$$\begin{array}{r} 10100 \rightarrow 01011 \\ + \quad \quad 1 \\ \hline 01100 \end{array}$$

$$\begin{array}{r} 10011 \\ + 01100 \\ \hline 11111 \end{array}$$



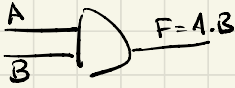
elde olsa sayı pozitif olur
elde yok sayı negatif sayının
ne olduğunu anlamak için tekrar
2's complement yapacağız



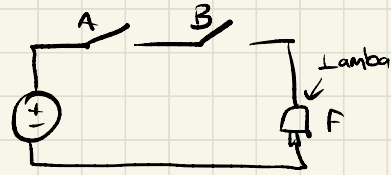
$$\begin{array}{r} 00000 \\ + \quad \quad 1 \\ \hline 00001 \end{array} \rightarrow \text{Sayı miç -1}$$

3 farklı kodu, Aiken, Grey,

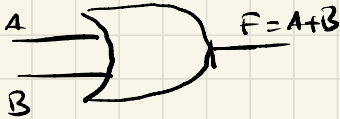
Ve (AND) işlemi



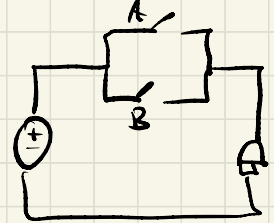
A	B	$F = A.B$
0	0	0
0	1	0
1	0	0
1	1	1



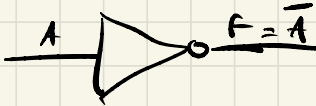
Ve ya (OR) işlemi



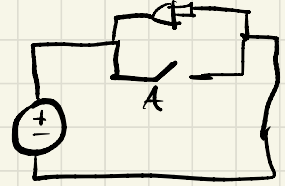
A	B	$F = A+B$
0	0	0
0	1	1
1	0	1
1	1	1



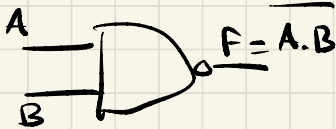
Tümlenme İşareti (NOT)



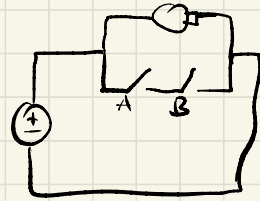
A	$F = \bar{A}$
0	1
1	0



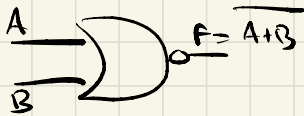
Tümlenme Ve İşlemi (NAND)



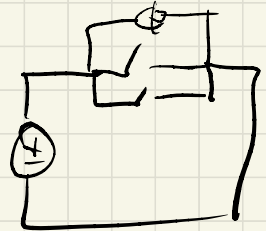
A	B	$F = \overline{A.B}$
0	0	1
0	1	1
1	0	1
1	1	0



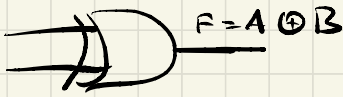
Tümlenme Ve ya İşlemi (NOR)



A	B	$F = \overline{A+B}$
0	0	1
0	1	0
1	0	0
1	1	0



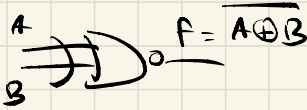
YA DA - ÖZEL VEYA (XOR)



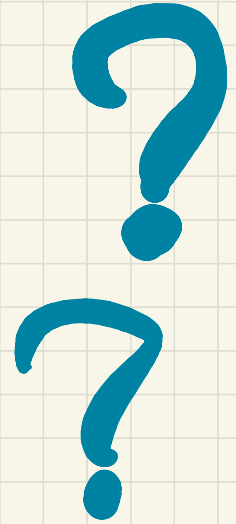
$$F = A \oplus B = \bar{A}B + A\bar{B}$$

A	B	$F = A \oplus B$
0	0	0
0	1	1
1	0	1
1	1	0

TYA DA - VEYA Degil (X NOR)



A	B	F
0	0	1
0	1	0
1	0	0
1	1	1



2. hafta Bitimi

3. hafta Başlangıcı

Slaytton devam et. H 012

Sayfa 26'ya kadar

ÖR: $(\bar{A} + ABC)(A + \bar{C}) = ?$

OR:

$$\overline{A}\overline{B}\overline{C} + A(C\overline{D} + \overline{B}) = ?$$

$$\overline{A}\overline{B}\overline{C} + A\overline{C}\overline{D} + A\overline{B}$$

$$\overline{B}(A + \overline{A}\overline{C}) + A\overline{C}\overline{D}$$

$$\overline{B}(A + \overline{C}) + A\overline{C}\overline{D}$$

OR: $(\overline{A}\overline{B} + C)(\overline{B}\overline{D} + \overline{C}\overline{E}) + (\overline{A}\overline{B} + C)$

$$(x + \overline{x})\overline{x} +$$

$$\cdot (\overline{A}\overline{B} + C) + (\overline{B}\overline{D} + \overline{C}\overline{E})$$
$$(\overline{A} + \overline{B} \cdot \overline{C}) (\overline{A} + \overline{B} + C)$$
$$(\overline{A}\overline{B} + \overline{A}\overline{C})$$

Min term - Max term

Sum of Product

X	Y	Z	F
0	0	0	0 $\rightarrow x+y+z$
0	0	1	1 $\rightarrow \overline{x}\overline{y}z$
0	1	0	0 $\rightarrow x + \overline{y} + z$
0	1	1	1 $\rightarrow \overline{x}yz$
1	0	0	0 $\rightarrow \overline{x} + y + z$
1	0	1	0 $\rightarrow \overline{x} + \overline{y} + \overline{z}$
1	1	0	1 $\rightarrow x\overline{y}\overline{z}$
1	1	1	1 $\rightarrow xyz$

$$f(x, y, z) = m_1 + m_3 + m_6 + m_7$$
$$= \overline{x}\overline{y}z + \overline{x}yz + x\overline{y}\overline{z} + xyz$$

$$F(x, y, z) = M_0 \cdot M_2 \cdot M_4 \cdot M_5$$
$$= (x+y+z)(x+\overline{y}+z)(\overline{x}+y+z)(\overline{x}+\overline{y}+\overline{z})$$

Product of Sum

ÖR= $f(a,b,c) = ab + ac$

1-) Doğruluk Tablosu ?

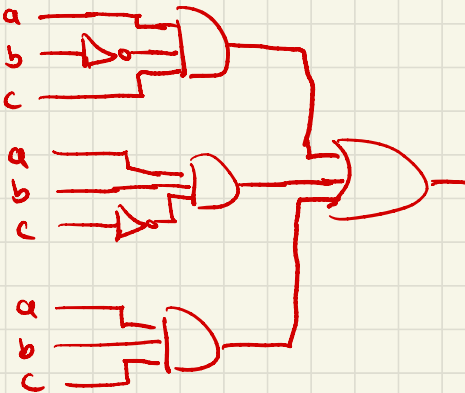
2-) min ve max terimleri yazılacak

3-) Lojik kapılarda minterm ve maxterm tasarımları yapılacak.

a	b	c	f
0	0	0	0 $\rightarrow a+b+c$
0	0	1	0 $\rightarrow a+\underline{b}+\underline{c}$
0	1	0	0 $\rightarrow a+b+\underline{c}$
0	1	1	0 $\rightarrow a+\underline{b}+\underline{c}$
1	0	0	0 $\rightarrow \underline{a}+b+\underline{c}$
1	0	1	1 $\rightarrow a\underline{b}c$
1	1	0	1 $\rightarrow a b \underline{c}$
1	1	1	1 $\rightarrow a b c$

$$f(a,b,c) = M_0.M_1.M_2.M_3.M_4$$

$$f(a,b,c) = m_5 + m_6 + m_7$$



$$\overline{OR} = f(a, b, c, d) = \bar{c}d(a+b) + cd(a+b) \\ = (a+b)d$$

a	b	c	d	F
0	0	0	0	0 → a+b+c+d
0	0	0	1	0 → a+b+c+ $\bar{d}$
0	0	1	0	0 → a+b+ $\bar{c}$ +d
0	0	1	1	0 → a+b+ $\bar{c}$ + $\bar{d}$
0	1	0	0	0 → a+b+c+d
0	1	0	1	1 → $\bar{a}\bar{b}\bar{c}d$
0	1	1	0	0 → a+b+c+d
0	1	1	1	1 → $\bar{a}bcd$
1	0	0	0	0 → a+b+c+d
1	0	0	1	1 → $a\bar{b}\bar{c}d$
1	0	1	0	0 → a+b+c+d
1	0	1	1	1 → $abcd$
1	1	0	0	0 → a+b+c+d
1	1	0	1	1 → $ab\bar{c}d$
1	1	1	0	0 → a+b+c+d
1	1	1	1	1 → $abcd$

$$f(a, b, c, d) = m_5 + m_7 + m_9 + m_{11} + m_{13} + m_{15}$$

$$f(a, b, c, d) = M_0 \cdot M_1 \cdot M_2 \cdot M_3 \cdot M_4 \cdot M_6 \cdot M_8 \\ \cdot M_{10} \cdot M_{12} \cdot M_{14}$$

$$\bar{c}d(a+b) + cd(a+b)$$

$$(b\bar{b})\bar{a}\bar{c}d + b\bar{c}d + acd + bcd$$

$$a\bar{b}\bar{c}d + a\bar{b}cd + \bar{a}b\bar{c}d + \bar{a}bcd + abcd + \bar{a}bcd$$

Literal → L = kaç değişken kullanıldığı yazılır

Gate → kapıların kaç girişle kullanıldığı

GN → " " " " " ve not kapılarını ekle

$$F = ABC + \bar{A}\bar{B}\bar{C}$$

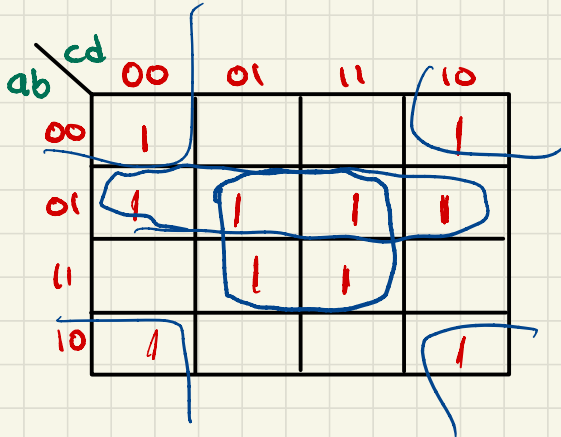
$$L = 6$$

$$G = 8$$

$$GN = 11$$

~ Karnaugh Maps ~

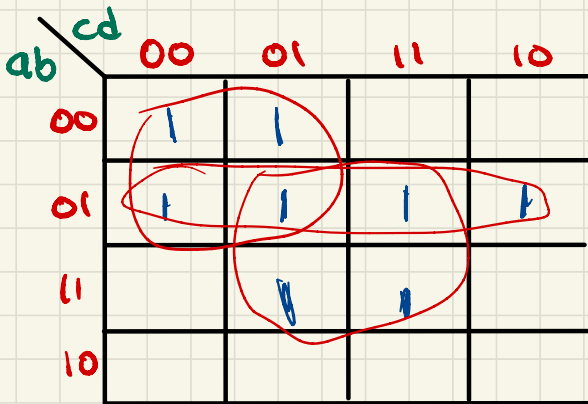
AYRIK Matematik ten baki.



$$x'z' + w'x + xz$$

$$b'd' + ab + bd$$

✓

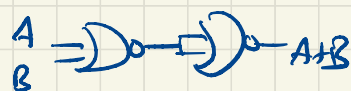
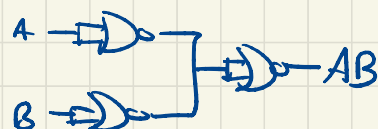
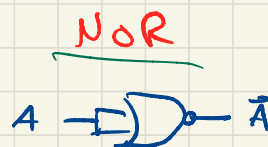
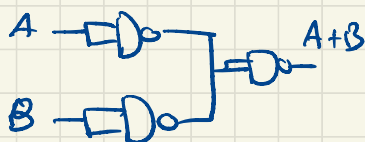
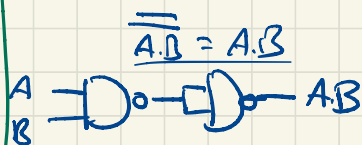
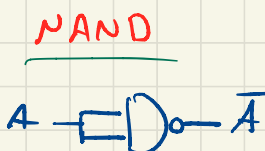
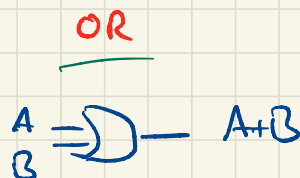
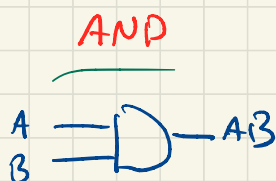
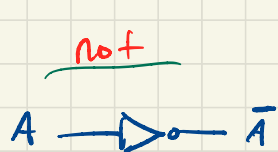


$$\bar{a}\bar{c} + bd + \bar{a}b$$

ab \ cd				
	00	01	11	10
00	X	1	1	X
01	0	1	1	0
11	0	0	0	0
10	1	0	0	X

$$a'd + b'd'$$

● → maxterm



OR

$$f(a,b,c) = \bar{a}\bar{b}c + \bar{a}bc + a\bar{b}c + abc$$

① → indirgeniz K-Map

② → lojik diyagramı NAND ile gerçekleştiriniz.

③ → " " NOR " "

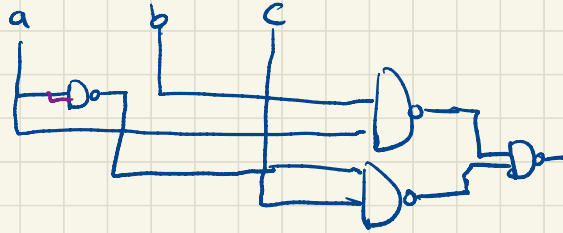
① →

a \ bc	00	01	11	10
0		1	1	
1			1	1

$$f = \underline{\underline{\bar{a}c + ab}} \rightarrow \overline{(\bar{a}c) + (ab)} \\ \rightarrow \overline{(\bar{a}c)} \cdot \overline{(ab)} \\ \bar{f} = f$$

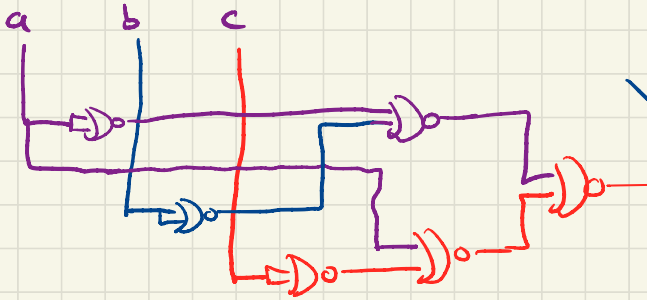
② →

$$\overline{(\bar{a}c) \cdot (ab)}$$



③

$$(a + \bar{c}) \cdot (\bar{a} + \bar{b}) = \overline{(a + \bar{c}) + (\bar{a} + \bar{b})}$$



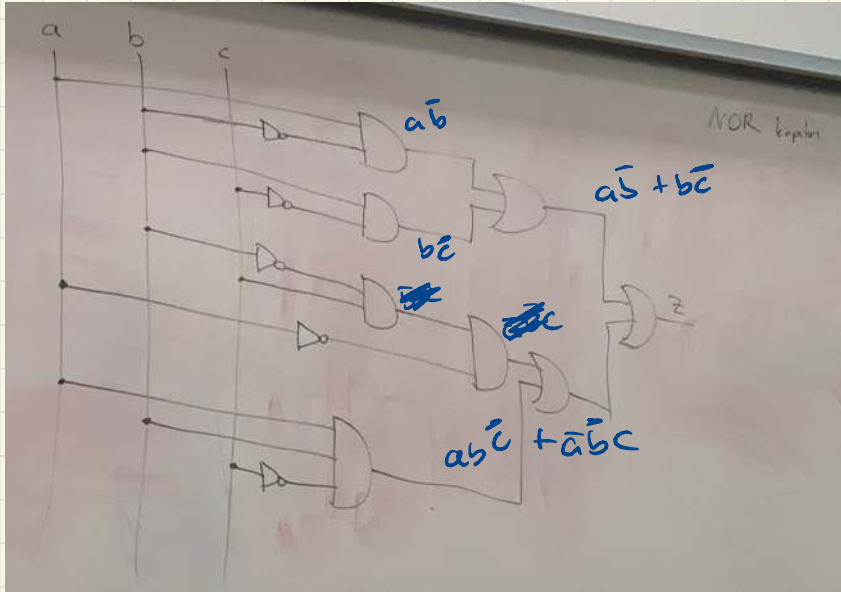
Yanlış olabilir!

ÖR

$$f(a, b, c) = (a + b)(\bar{a} + c)$$

① → Lojik diyagramını NOR ile gerçekleştiriniz

$$\bar{f} \rightarrow \overline{(a + b) + (\bar{a} + c)}$$



$$F = ?$$

$$F(a, b, c) = a\bar{b}c + a\bar{b}\bar{c} + a\bar{b}c + a\bar{b}\bar{c}$$

ÖR

a	b	c	F
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	1
1	0	0	0
1	0	1	1
1	1	0	1
1	1	1	1

① SOP şeklinde z fonksiyonunu (kard)

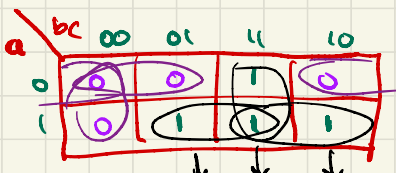
② NAND

③ NOR

④ eleman karmaşıklığı açısından hangi tür tasarımın kullanılması gerektiğini kaygılayınız.

$$\textcircled{a} \rightarrow F(a, b, c) = \bar{a}bc + a\bar{b}c + ab\bar{c} + abc$$

$$\bar{a}bc + a + b + c$$



$$\text{miniterm} \leftarrow ac + bc + ab$$

$$(a+b)(a+c)(b+c)$$

↓
maxterm

② → $\bar{F} = f$ $(\overline{a \cdot c})(\overline{b \cdot c})(\overline{a \cdot b}) \rightarrow \text{nand}$ için

③ → $\bar{F} = f$ $(\overline{a+b}) + (\overline{b+c}) + (\overline{a+c}) \rightarrow \text{nor}$ için

KOMBİNAZONSA L DEVRELER

SORU

Bir hava alanından şehir merkezine 20 dk'da bir helikopter sefari düzenlenmektedir. Helikopterlerde bulunan bir sayıcı değerli Yeterli Kodda helikoptere binen yolcuları saymaktadır. helikopterler 7 kişiliktir. İfeli kopterlerin kalkması için

- Kalkış zamanı henüz gelmediyse helikopterin dolmuş olması
 - " " " geldiye se , içinde en az 2 yolcu olması gerekmektedir
- Pilotun kalkış bilmesi için gerekli isareti verecek kombinazonson l lojik devreyi tasarlayınız.

if zone > 7 ✓ time don't care X
if zone >= 2 ✓ time care ✓

Input				Output
time	zone ₂	zone ₁	zone ₀	
0	0	0	0	0
0	0	0	1	0
0	0	1	0	0
0	0	1	1	0
1	0	0	0	0
1	0	0	1	0
1	0	1	0	0
1	0	1	1	0
1	1	0	0	0
1	1	0	1	0
1	1	1	0	0
1	1	1	1	0
1	1	1	1	1

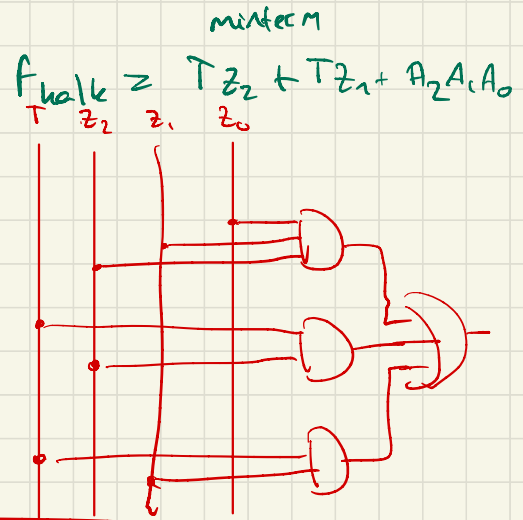
zone 27
time do not care

time has come
but zone < 2

time has come
and zone >= 2

Same

T	z_1	z_0	00	01	11	10
00						
01						
11						
10						



-(Toplama / Çıkarma) Aritmetik işlemler

- Yarı toplayıcı devre
- Tam toplayıcı devre
- Yarı çıkartıcı
- Tam çıkartıcı
- Tam Toplama / Çıkarma Devresi
- Seçiciler
- Kod Gözücüler (Decoder - DEC)
- Kodlayıcılar (Encoder - ENC)
- Dağıtıcılar (Demux - Demultiplexer)
- ALU

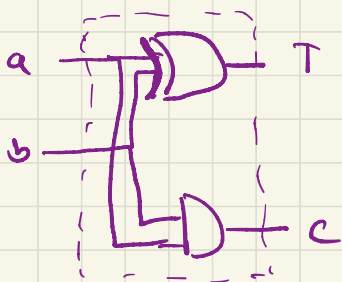
Yarı Toplayıcı

a	b	T	C
0	0	0	0
0	1	1	0
1	0	1	0
1	1	0	1

a	b	0	1
0	0	0	1
1	0	1	0

$$= T = a \oplus b$$

a	b	0	1
0	0	0	0
1	0	0	1

$$= C = a \cdot b$$


Tam Toplayıcı

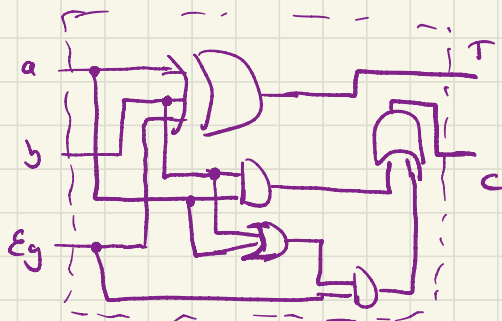
Eg	a	b	T	C
0	0	0	0	0
0	0	1	1	0
0	1	0	1	0
0	1	1	0	1
1	0	0	1	0
1	0	1	0	1
1	1	0	0	1
1	1	1	1	1

Eg	ab	00	01	11	10
0	0	0	1	0	1
1	1	1	0	1	0

$$T = Eg \oplus a \oplus b$$

Eg	ab	00	01	11	10
0	0			1	
1	1		1		1

$$C = ab + Eg(b + a)$$



Yarı Çıkarma

a	b	F	B
0	0	0	0
0	1	1	1
1	0	1	0
1	1	0	0

$$F = a \oplus b$$

$$B = \bar{a} \cdot b$$

$$\begin{array}{r} 30 \\ - 5 \\ \hline 25 \end{array}$$

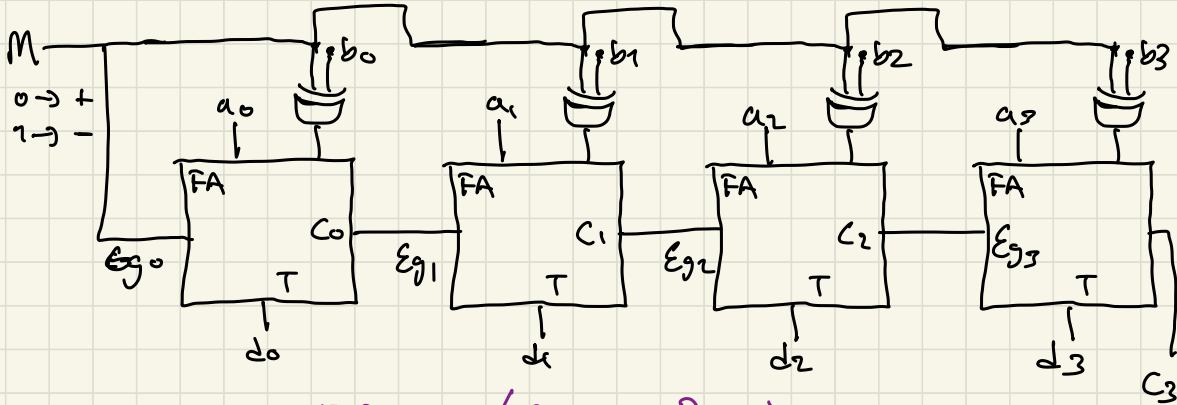
Tam Çıkarma

a	b	Eg	F	B
0	0	0	0	0
0	0	1	1	1
0	1	0	1	1
0	1	1	0	1
1	0	0	1	0
1	0	1	0	0
1	1	0	0	0
1	1	1	1	1

$$F = a \oplus b \oplus Eg$$

$$B = b \cdot Eg + \bar{a} Eg + \bar{a} b$$

$$B = bEg + \bar{a}(Eg + b)$$



TOPLAMA/Çıkarma Devresi

4 girişli 1 çıkışlı deneye doğal BCD kodunuoluşturmak için 2 veya 3 e ten bölünebilir ya da logic 1

a) $r.T = ?$

b) Çıkış ifadesini indirgeyiniz.

c) Nand ile gerçekleştirin

A_3	A_2	A_1	A_0	F
0	0	0	0	1
0	0	0	1	0
0	0	1	0	1
0	0	1	1	1
0	1	0	0	1
0	1	0	1	0
0	1	1	0	1
0	1	1	1	0
1	0	0	0	1
1	0	0	1	1

$$F = \bar{A}_3 \bar{A}_2 \bar{A}_1 \bar{A}_0 + \bar{A}_3 \bar{A}_2 \bar{A}_1 A_0 + \bar{A}_3 \bar{A}_2 A_1 \bar{A}_0 + \bar{A}_3 \bar{A}_2 A_1 A_0 + \bar{A}_3 A_2 \bar{A}_1 \bar{A}_0 + \bar{A}_3 A_2 \bar{A}_1 A_0 + \bar{A}_3 A_2 A_1 \bar{A}_0 + \bar{A}_3 A_2 A_1 A_0$$

$$(A_3 + A_2 + A_1 + \bar{A}_0) (A_3 \bar{A}_2 + A_1 \bar{A}_0) (A_3 + \bar{A}_2 + \bar{A}_1 + \bar{A}_0)$$

$$\bar{P} = \overline{A_0 + A_3 + \bar{A}_2 A_1}$$

$$(A_0 \bar{A}_3) \cdot (\bar{A}_2 A_1)$$

$A_3 A_2$	$A_1 A_0$ 00	01	11	10
00	1	0	1	1
01	1	0	0	1
11	x	x	x	x
10	1	1	x	x

$$F = \bar{A}_0 + A_3 + \bar{A}_2 A_1$$

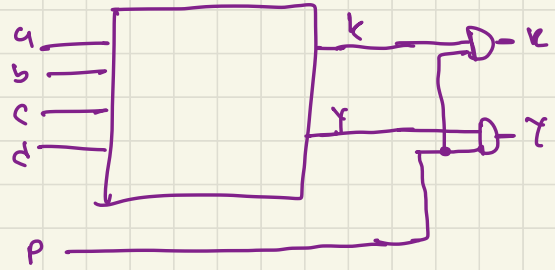
Bir aynaya katılan 4 kişi kararlarını önlerinde bulunan butona basarak belirtiyorlar. Teklifi kabul eden kişi butona basmaktadır. Salonu bulunan yeşil ve kırmızı olan 2 lamba ile

kabul = 0 veya 1 ise → kırmızı
 " 2 2 " → her ikisi
 " = 3 " 4 " → yeşil

a) T.T = ?

b) Σ deveyi!

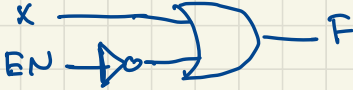
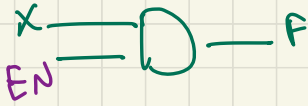
P	a	b	c	d	K	Y
0	0	0	0	0	1	0
1	0	0	0	1	1	0
2	0	0	1	0	1	0
3	0	0	1	1	1	0
4	0	1	0	0	1	0
5	0	1	0	1	1	0
6	0	1	1	0	1	0
7	0	1	1	1	1	0
8	1	0	0	0	1	0
9	1	0	0	1	1	0
10	1	0	1	0	1	0
11	1	0	1	1	1	0
12	1	1	0	0	1	0
13	1	1	0	1	1	0
14	1	1	1	0	1	0
15	1	1	1	1	1	0



Yeşil

cd \ ab	00	01	11	10
00	0	0	1	0
01	0	1	1	1
11	1	1	1	1
10	0	1	1	1

Enabling function



$EN \rightarrow 0$ ise $F \rightarrow 0$
 $EN \rightarrow 1$ ise $F \rightarrow X$

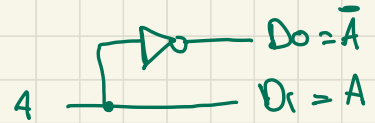
$EN \rightarrow 0$ ise $F \rightarrow 1$
 $EN \rightarrow 1$ ise $F \rightarrow X$

Decoding

$n = \text{input} > \text{ise} \rightarrow n \leq m \leq 2^n$
 $m = \text{output}$

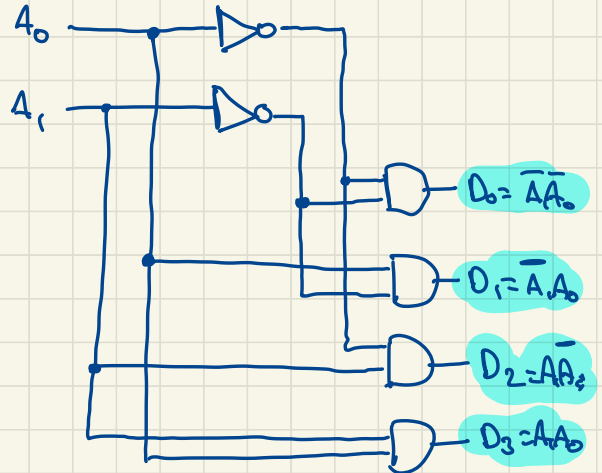
1-to-2 Line Decoder

A	D_0	D_1
0	1	0
1	0	1

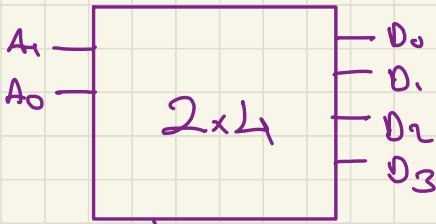


2-to-4 - Line Decoder

A_1	A_0	D_0	D_1	D_2	D_3
0	0	1	0	0	0
0	1	0	1	0	0
1	0	0	0	1	0
1	1	0	0	0	1

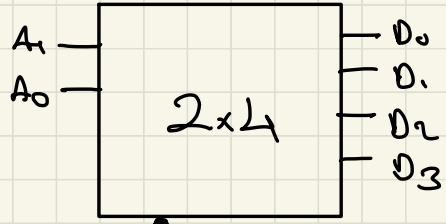


Decoder With Enable



$E_n = 1$ de aktif

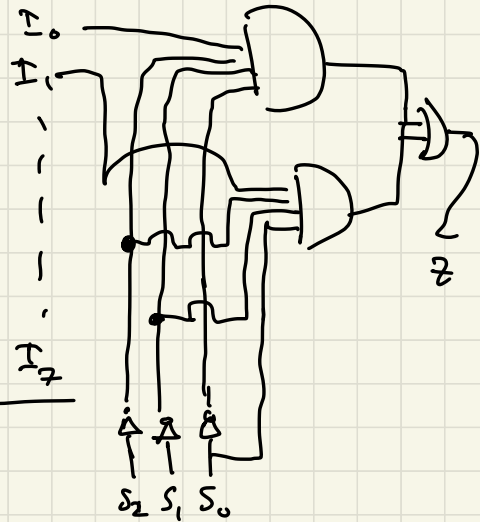
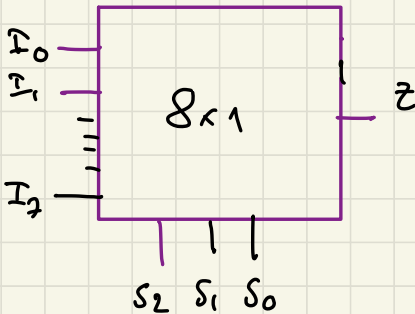
E_n	A_1	A_0	D_0	D_1	D_2	D_3
0	X	X	0	0	0	0
1	0	0	1	0	0	0
1	0	1	0	1	0	0
1	1	0	0	0	1	0
1	1	1	0	0	0	1



$E_n = 0$ de aktif

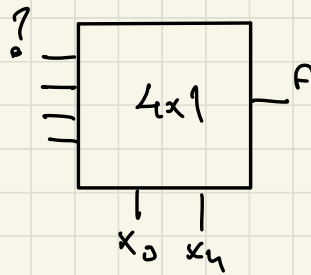
E_n	A_1	A_0	D_0	D_1	D_2	D_3
1	X	X	0	0	0	0
0	0	0	1	0	0	0
0	0	1	0	1	0	0
0	1	0	0	0	1	0
0	1	1	0	0	0	1

MUX



QR

x_1	x_2	x_3	x_4	z
0	0	0	0	0
0	0	0	1	0
0	0	1	0	1
0	0	1	1	1
0	1	0	0	0
0	1	0	1	1
0	1	1	0	0
0	1	1	1	1
1	0	0	0	1
1	0	0	1	1
1	0	1	0	1
1	0	1	1	1
1	1	0	0	0
1	1	0	1	0
1	1	1	0	0
1	1	1	1	0

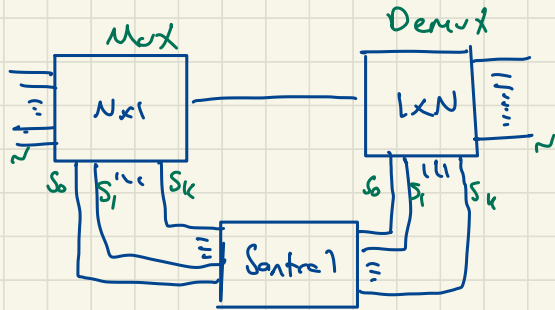
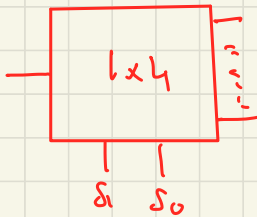


$$f = \overline{x_1} \overline{x_2} x_3 + \overline{x_1} x_2 x_4 + x_1 \overline{x_2}$$

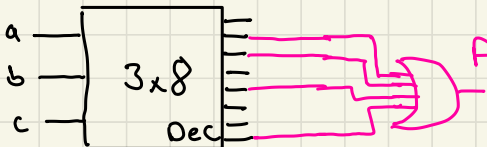
$$f = \overline{x_3} \overline{x_4} (\overline{x_1} \overline{x_2}) + \overline{x_3} x_4 (\overline{x_1} x_2 + x_1 \overline{x_2}) + x_3 \overline{x_4} (\overline{x_1} \overline{x_2} + x_1 \overline{x_2}) + x_3 x_4 (\overline{x_1} + \overline{x_2})$$

i	x_3	x_4	I_i
0	0	0	$\overline{x_1} \overline{x_2}$
1	0	1	$\overline{x_1} \oplus \overline{x_2}$
2	1	0	$\overline{x_2}$
3	1	1	$\overline{x_1} + \overline{x_2}$

- Digitaler (Demultiplexer - Demux)

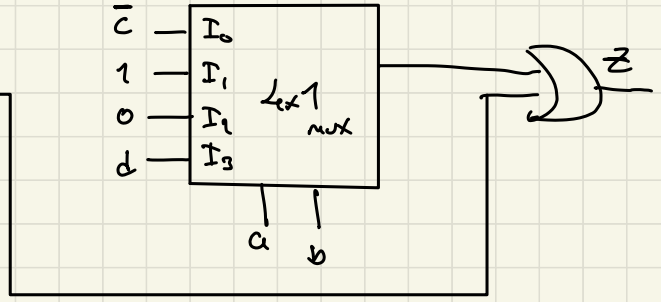
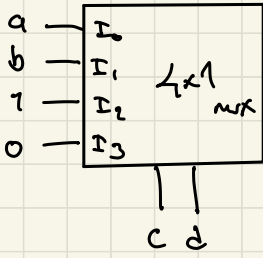


OR $f(c,b,a) = \sum(1,2,4,7)$



ÖR

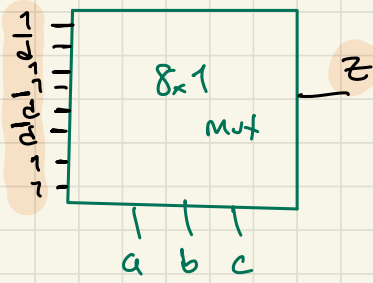
Soru = Doğruluk tablosunu oluşturunuz.



a	b	c	d	z
0	0	0	0	1
0	0	0	1	1
0	0	1	0	1
0	0	1	1	0
0	1	0	0	1
0	1	0	1	1
0	1	1	0	1
0	1	1	1	1
1	0	0	0	1
1	0	0	1	0
1	0	1	0	1
1	0	1	1	0
1	1	0	0	1
1	1	0	1	1
1	1	1	0	1
1	1	1	1	1

ÖR

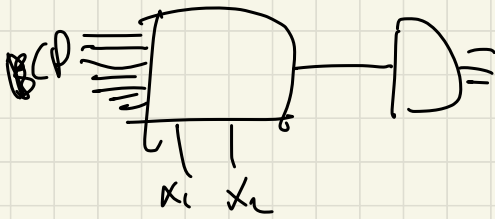
a, b, c seçme kolı ile 8x1 bir mux yapın



BCD aktarması
(0-9)

bir de eşlik biti var

BCD deki 1'leri tek
 $x_1 x_2$
yapan
eşlik biti
Seçici mux ile görebiz



x_4	x_3	x_2	x_1	z
0	0	0	0	1
0	0	0	1	0
0	0	1	0	0
0	0	1	1	1
0	1	0	0	0
0	1	0	1	1
0	1	1	0	1
0	1	1	1	0
1	0	0	0	0
1	0	0	1	1
1	0	1	0	0
1	0	1	1	0
1	1	0	0	1
1	1	0	1	0
1	1	1	0	0
1	1	1	1	1

$\overline{x_2}$ and $\overline{x_3}$
 x_4 or x_3

